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Analyzing and Synthetizing

Human Computer Interaction

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Hall of Fame or Shame?

User Needs Edition

Beware: we are missing the general context here!

is a solution, not a need (is not a verb)

- Users need a faster horse

The verb is not a verb, but a sentence (not a need)

- Users need to have financial help

It is a user need

- Users needs a way to move faster from one place to another

not so specific; the verb is not a real verb (not a user need)

- Users need to have more tools

It is a user need

- Users need to practice more with the appropriate tools

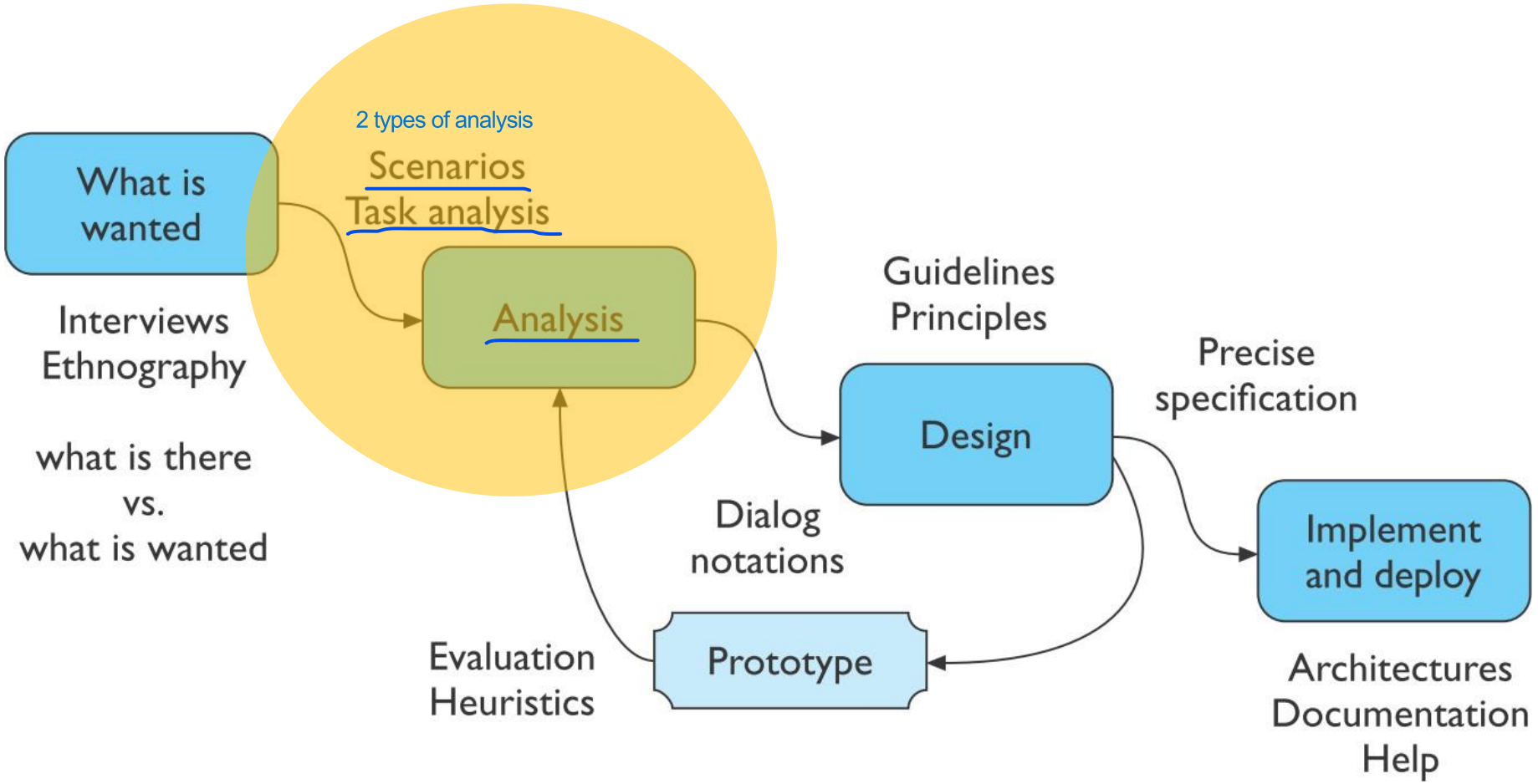
Is mix of Fame and Shame (it isn't a need; it is what we want to achieve)

- Users need to be able to run faster

From Assignment 1:

"Needs are human emotional or physical necessities. […]
Needs are verbs (activities and desires with which your user could use help), not nouns (solutions). […]
It can be helpful to use the phrases 'needs a way to' or 'needs to be able to' in your list of user needs."

Human-Centered Design Process



Goals of the users

- Create **design goals**
  - As an intermediate representation before the user interface design
- Make the user needs' analysis **explicit** From the need we need to understand the Goals
- Think about the *interplay* between the activity that someone has and the interface we offer
- **Represent** and synthetize the results of the analysis and the design goals



Tasks and Their Analysis

How people perform their activities

Tasks

- Task: the structured **set of activities**/high-level actions required to **achieve** a user goal
  - It says what a person *wants to do*, not how, while describing a *complete* goal (ci aiutano a capire il goal completo)

Questo perchè le attività di un task non sono nel dettaglio, ma sono high-level

- Often, given a domain, you have a **mix** of tasks with different **complexity**
  - Simple tasks – common or introductory (ogni persona lo fa: fa parte dei need di cui tutti gli utenti hanno bisogno)
  - Moderate tasks
  - Complex tasks – infrequent or for power/extreme users
    - Extreme users can be described as the people on either end of the spectrum of users of a product or service

Task Analysis

See artifacts like all the things that are tangible

- Task analysis is the study of the way people perform their activities

Task analysis help

- ~~Aim is~~ to determine:
  - what they **do** (steps) per arrivare ad un certo goal (cioè cosa fa l'utente per arrivare ad un certo goal)
  - what things they **use** (artifacts) (cioè quali device vengono usati dagli utenti)
  - how well they **succeed** (goals, pain points) (quanto efficace ed efficiente il design per portare a termine il goal)

Sample Task: To Clean The House (I)

- **Steps:** What they are going to do?
  - get the vacuum cleaner out
  - fix the appropriate attachments
  - clean the rooms e.g. that is very high level these activities
  - when the dust bag gets full, empty it
  - put the vacuum cleaner and tools away
- Must know and use different **artifacts:**
  - vacuum cleaners, their attachments, dust bags
  - cupboards, rooms
  - …

Sample Task: To Clean The House (II)

- **Goals:**
  - Here your *point of view* comes in
  - Removing dust? -> narrow goal
  - Tidying up the house after a party?
  - Hosting people for the dinner?
  - Having a satisfying evening? -> wide goal

Simple task

Goal is the same, but done in different ways (depend on context and final goal)

Sample Task: To Clean The House (III)

- **Pain points:** (something that happens sometime, not often)
  - Narrow version: Why I need to empty the dust bag?
  - Broader version: Why I need a vacuum cleaner to have the house cleaned up? We can use different type of artifacts

What is a Tasks?

- «A **task** is a **goal** together with some ordered set of **actions**.» (Benyon)

Goal

- A state of the application domain that a work system (user+technology) wishes to achieve.
- Specified at particular levels of abstraction. (can be narrow or wide or specific based on context)

Task

- A structured set of activities required, used, or believed to be necessary by an agent (human, machine) to achieve a goal using a particular technology.
- The task is broken down into more and more detailed levels of description until it is defined in terms of actions.

Inside the task, you have different actions

Action

- An action is a task that has no problem solving associated with it and which does not include any control structure. but you still define how the task is compute (task is structured and composed by a set of activities/actions)
- Actions are 'simple tasks'.

What You Learn with Task Analysis

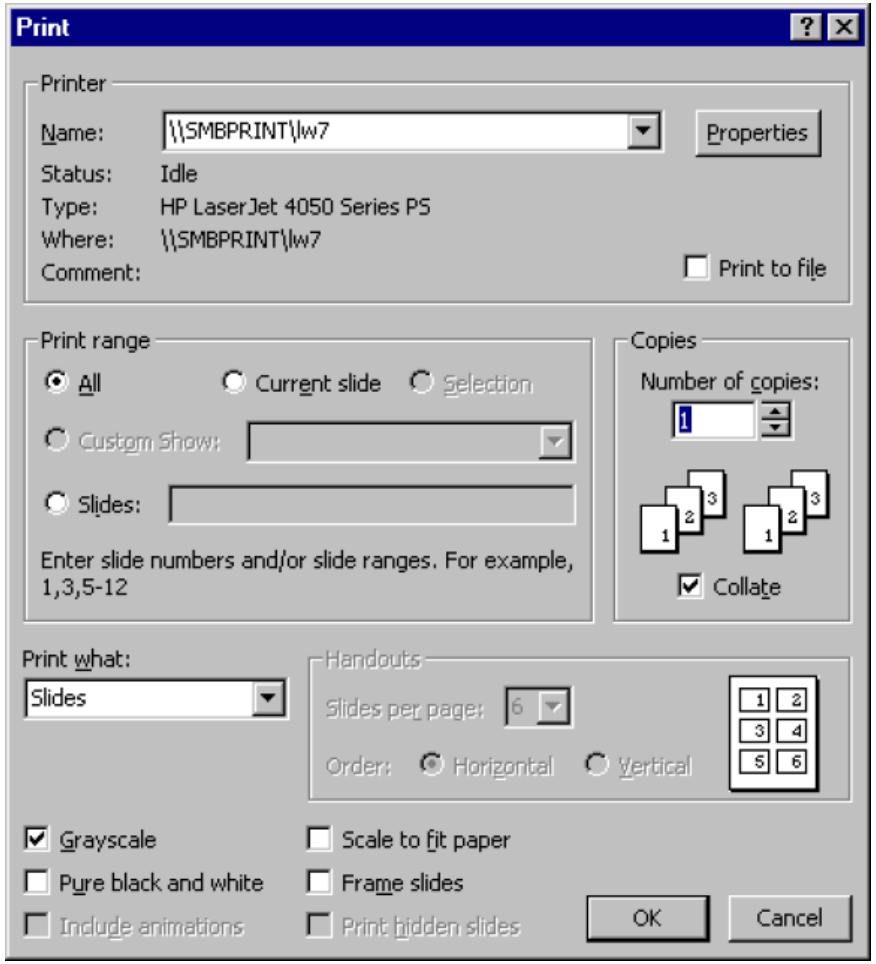
- What your users' goals can be; what they are trying to achieve
- What users actually do to achieve those goals (the set of activities)
- What experiences (personal, social, and cultural) users bring to the tasks
- How users are influenced by their physical environment
- How users' previous knowledge and experience influence:
  - How they think about their work
  - The workflow they follow to perform their tasks
  - The pain points they experience to perform the tasks

experiences and environment is about the context in which the user do a task

Why Is It Useful?

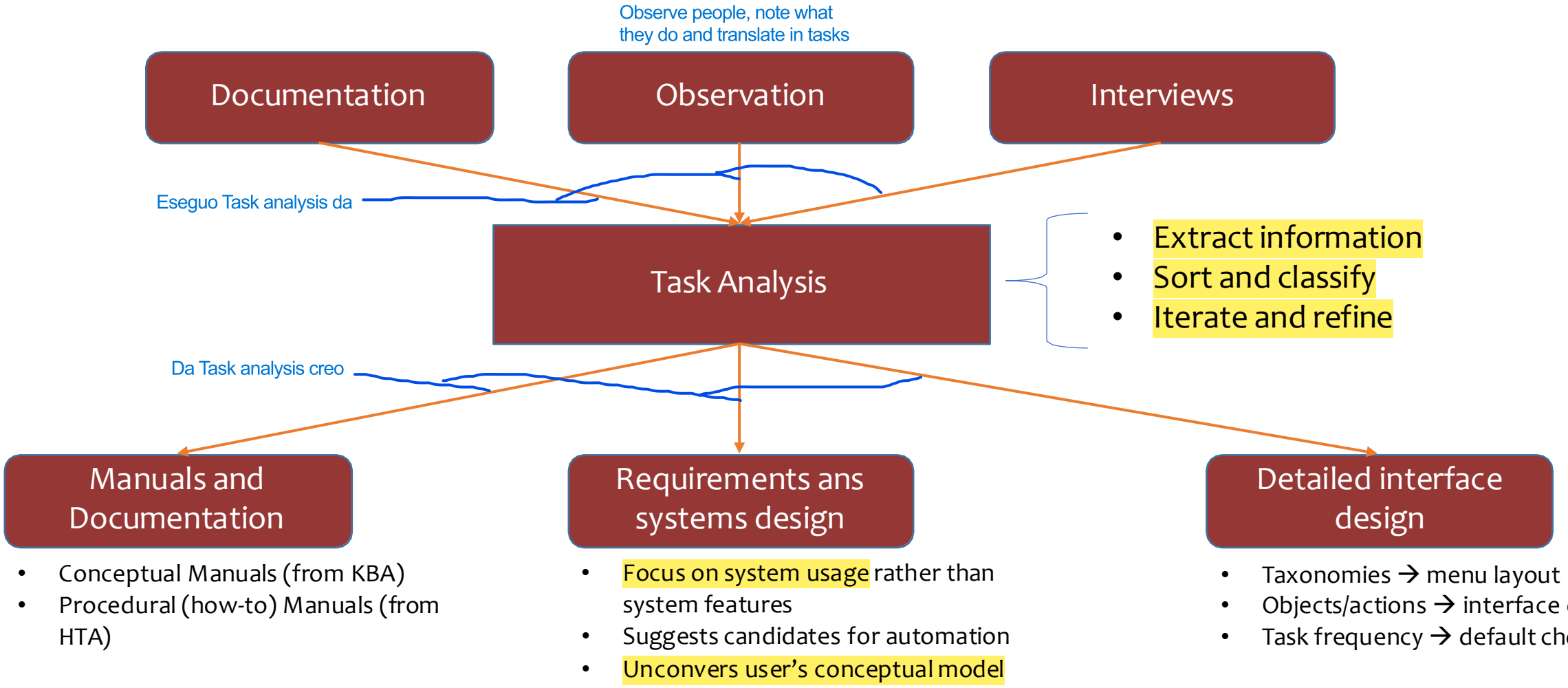
- Task analysis is the process of learning about ordinary users by observing them in action to understand in detail **how** they perform their **tasks** and achieve their intended **goals**
- Tasks analysis helps in:
  - Identifying the tasks that your application **must support**
  - Refining or re-defining your app's navigation or search
  - Application requirements gathering
  - Developing your content strategy and app structure
  - The initial stages of Prototyping
  - Performing usability testing (you put the user in a specific task and see how the user react)

Example



- Tasks are used to plan for the layout of the application window
- Proximity and Boundaries reflect the decomposition of tasks
- Order of tasks is not mandatory

Where It Fits



- Conceptual Manuals (from KBA)
- Procedural (how-to) Manuals (from HTA)

- Focus on system usage rather than system features
- Suggests candidates for automation
- Unconvers user's conceptual model

- Taxonomies → menu layout
- Objects/actions → interface objects
- Task frequency → default choices

Characteristics of Task Analysis

- Task analysis is easier when you have well-defined **workflows** (so sequence of steps to do Task analysis) (e.g., planning a trip somewhere)
  - or **repeated activities**, such as scheduling
- Challenge:
  - We **do not** design tasks, ~~but~~ and then interfaces — We start from Task and then we design the interfaces
  - Tasks and ~~objects~~ interfaces do not map 1:1 — complex tasks can have multiple interfaces
    - e.g., a web app has multiple tasks
  - People use the same interface and application to achieve slightly different results or do things differently one another



Noi faremo: da needfinding, define the tasks
(goal of the tasks and set of activities for each)

[Some] Techniques for Task Analysis

Varie tecniche che noi non vediamo

- Task decomposition – Splitting tasks into sub-tasks and their ordering
- Knowledge-based techniques – Any information and instructions that users need to know, and how that knowledge is organized
- Entity-relationship-based analysis – Identify actors, objects, relationships and their actions
- Ethnography – Observation of users' behavior in the use context
- Protocol analysis – Observation and documentation of actions of the user. This is achieved by authenticating the user's thinking. The user is made to think aloud so that the user's mental logic can be understood.

Sketches

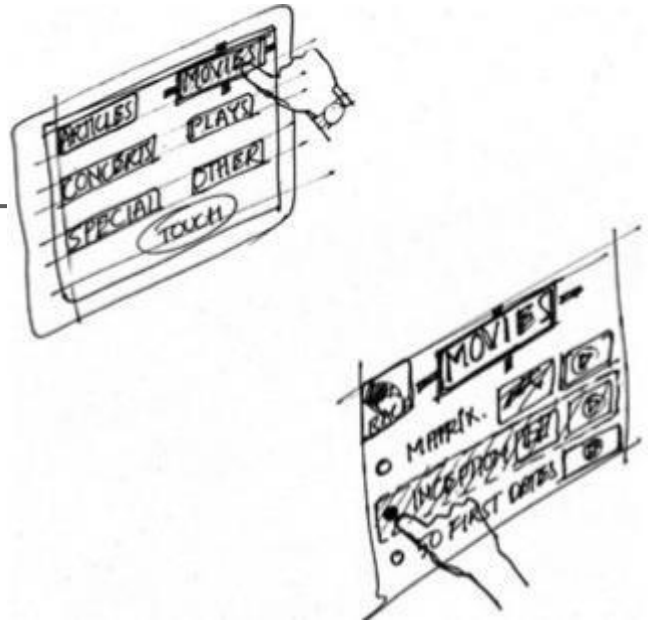
Quick drawings to convey a part of the interface, or a feeling about a device

Manual drawing phisically or virtually

After task analysis, we need to synthetize and visualise the interaction (not the interface: quella che disegniamo può non essere l'interfaccia finale dell'app)

Sketch

With Sketches you start to visualise the interaction



- An individual drawing showing
  - A single user interface screen (not the final one)
  - A drawing of an artifact part of the system
  - The shape of an interaction object

- Gives a static view of a possible interaction
- Helps setting the interaction context
- Often, part of a longer representation (e.g., a storyboard)



One way to do Scenario is Storyboard

Scenarios

can put together different sketches to get a Scenario

Possible sequences of actions for reaching user goals

Scenario

- Scenarios are stories for design: rich stories of interaction
- Description of how the user engages the interactive system to solve a specific task
- Formats:
  - Written summary, Use Case
  - Graphical sketches (→ Storyboard)
  - Flowcharts, Transition Diagram…

The interface is not so important. It is very important the interaction visualisation (how the user interact with the device)

needfinding -> user needs -> goals -> task -> set of activities for each task -> scenario

Level of Details In Scenarios

- Concrete Scenarios
  - Used for envisioning ideas and evaluation
  - One possible solution to a Conceptual Scenario (may try many alternatives) (e.g different devices: more scenarios)
  - Shows how technologies are used in the user context
  - Key design features are included
- Use Cases
  - Used for specification and implementation (→ software engineering)
- **Stories**
  - From needfinding
  - Used for understanding what people do and what they want
- **Conceptual** (abstract) **Scenarios** Focus on user interactions (not interfaces)
  - Used for generating ideas and specifying requirements
  - Abstracts tasks from stories
  - No reference to technology (tech means e.g. put javascript on a particular device)
  - May lead to different concrete scenarios

Storyboards

is a way to do scenarios

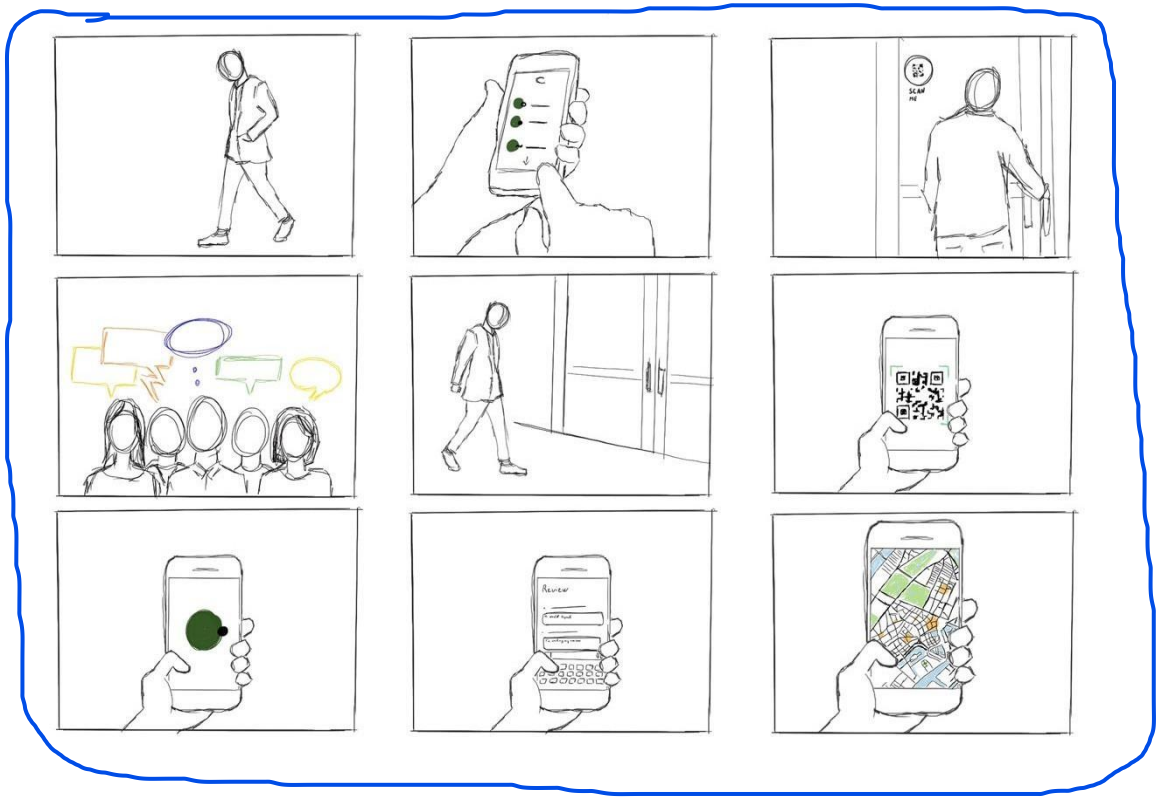
Comic book – like representation of user scenarios, with emphasis of how the system supports users in the development of the task

Storyboard

You can include text, if you want

- "A graphical depiction of the outward appearance of the intended system, without any accompanying system functionality"
- A hand-drawn comic that features the execution of a task (like a concrete scenario)
- With a few panels (sequence of sketches) it conveys what a person may accomplish
  - Always include people
- They communicate **flow**, showing what happens **at key points** in time
- No artistic skills are required
  - Not about "nice pictures"
  - About communicating ideas

Important here: One specific solution for a specific task



What To Find In a Storyboard

- Illustrate a goal (for the task) one storyboard = one task
- How a task unfolds (people interacting among themselves and with devices)
  - Repeated for all significant steps
- At the end, how they accomplish their goals (satisfactory outcome)

no illustrate errors

- Storyboards are **all about tasks**

Example

Doesn't need to be good looking, but understandable



This storyboard illustrates how the app had already downloaded the daily recipe to the user's smartphone, so he could look it up and check the shopping list while on the underground, before shopping for ingredients and making a healthy meal.

http://alexmevissen.com/2014/07/16/storyboarding/

Example



This storyboard illustrates how the app can show the user that a home cooked meal can be quicker than ordering food delivery, using left over ingredients in the fridge.

http://alexmevissen.com/2014/07/16/storyboarding/

Storyboards Should Convey…

- Setting (illustrate how the user interact with the system)
  - People involved
  - Environment
  - Task being accomplished
- Sequence (set of activities illustrated)
  - What steps are involved? (set of activities)
    - Not the detailed UI
    - What role the UI plays in helping users achieve their goal?
  - What leads someone to use the system?
    - The "trigger" for the task
  - What task is being illustrated?
- Satisfaction
  - What's the motivation for the user?
    - The end point to reach after all the steps
  - What's the end result?
  - What need are you "satisfying"?

Handling Dynamicity In Storyboards

Three technics to do Storyboards

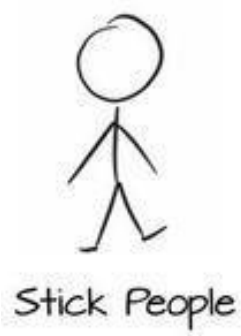
- Traditional storyboarding l'unico che useremo
  - "Comic book" conventions: actors, speech bubbles, background
  - Notes attached to each scene explaining what is happening
- Scored storyboards
  - When the user interface is highly dynamic, or contains specific media elements
  - Add specific annotations focusing on movement, colors, sounds, …
- Text-only storyboards
  - When the interaction behavior is too complex to compact into an illustration, use a longer text description

Why Hand-drawn?

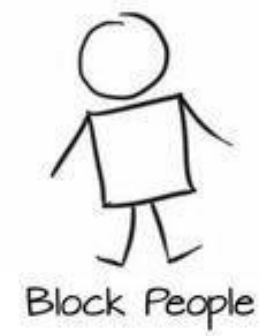
- Quick
  - No need to spend time in graphics tools (they would "push" you to focus on details, too
  - Able to experiment different scenarios
- Imprecise
  - Users feel free to express more comments and suggestions w.r.t. a more "polished" version
  - Focus on the content (the graphics is obviously ignored)
  - No distraction by fonts, colors, icons, …

~~Drawing~~ Sketching People

Puoi farli in qualsiasi modo



Stick People



Block People



Blob People



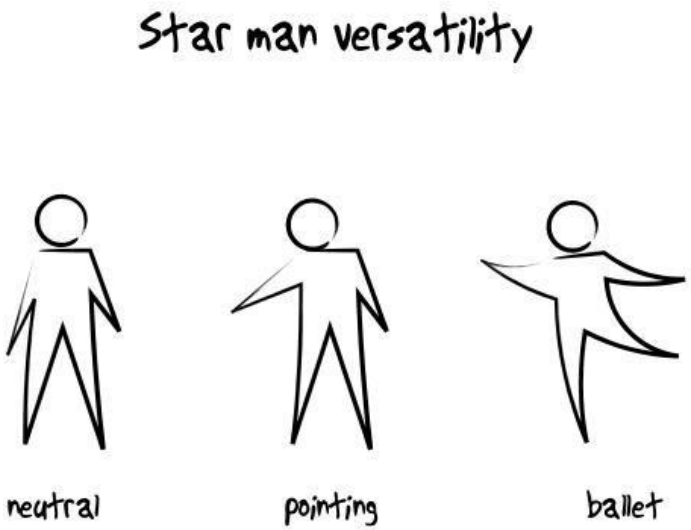
Star People



Triangle People



Use Your Imagination





You gonna do different storyboards, then you will select some and then you start to implement/prototype it

Benefits of Storyboards

- Emphasize how an interface accomplishes a task (Focus on interactions: Task, goal, activities, final goal)
- Focus the conversation and feedback on user tasks
- Get everyone on same page about the app's goals
- Avoid nitpicking about user interface details (buttons, etc.)



References

- Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale: Human Computer Interaction, 3rd Edition, Chapter 15 "Task Analysis"
- David Benyon: Designing Interactive Systems, Chapter 11 "Task Analysis"
- http://www.usabilitybok.org/task-analysis
- https://www.usability.gov/how-to-and-tools/methods/task-analysis.html

Acknowledgements

- Some icons from https://icons8.com



- Some material by
  - http://www.inf.ed.ac.uk/teaching/courses/hci/0708/lecs/tasks.pdf
  - https://www.tutorialspoint.com/human_computer_interface/design_process_and_task_analysis.htm
  - https://www.slideshare.net/alanjohndix/hci-3e-ch-15-task-analysis

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